

# Jiayi Chen

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Google Scholar | LinkedIn | GitHub | Homepage | ORCID

**Research Interests:** Embodied AI, vision-language navigation, LLM/VLM planning, autonomous driving.

**Technical Skills:** Python, C++, PyTorch, ROS, CARLA, Apollo, MPC, A\*, Sionna RT, Taichi RT.

## EDUCATION

### The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen)

Sept. 2024 – Present

*Ph.D. in Computer and Information Engineering, School of Science and Engineering (SSE)*

Shenzhen, China

- **Research Area:** Embodied Intelligence
- **Advisor:** Prof. Guangxu Zhu; **Co-advisor:** Prof. Shuai Wang

### Beijing University of Posts and Telecommunications (BUPT)

Sept. 2020 – Jun. 2024

*B.Eng. in Internet of Things Engineering, Intl. School; QMUL-BUPT Joint Programme*

Beijing, China

- **Weighted Average Mark:** 90.07/100; **GPA:** 3.73/4.0; ranked **6/189**
- Selected into the YePeida Institute Pilot Class, admitting approximately the top **7%** of students

## SELECTED PUBLICATIONS

**Jiayi Chen**, et al. “Bridging Large-Model Reasoning and Real-Time Control via Agentic Fast–Slow Planning.” *ICRA 2026*, accepted. [Paper] [Code]

**Jiayi Chen**, et al. “Opportunistic Collaborative Planning with Large Vision Model Guided Control and Joint Query-Service Optimization.” *IROS 2025*, pp. 951–958. [Paper]

**Jiayi Chen**, et al. “Not All Uncertainty Matters: Simulation-in-the-Loop Fast–Slow Reasoning for Decision-Critical Autonomous Driving System.” *IEEE TMC*, under review. [Code]

**Jiayi Chen**, et al. “The Universal Language of CSI: Unifying Wireless Sensing Across Devices and Environments.” *IEEE Communications Magazine*, accepted. [Paper] [Code]

## RESEARCH EXPERIENCE

### Embodied Fast–Slow Planning and Autonomous Driving

Jan. 2025 – Present

*Research lead; CUHK-Shenzhen*

Shenzhen, China

- Proposed **AFSP**, a VLM/LLM-to-semantic-A\* framework that decouples slow reasoning from real-time MPC control; CARLA evaluation reduced route completion time by over **12%** and maximum lateral deviation by up to **45%** versus MPC and A\*-MPC
- Developed **OCP** for decision-value-triggered cloud collaboration in open-world driving; reduced service calls by **50%**, improved success rate by over **6%**, and cut dynamic-scenario time by **26.2%**

### Foundation Models for Wireless Sensing

Feb. 2025 – Dec. 2025

*Research lead; accepted by IEEE Communications Magazine*

Shenzhen, China

- Proposed **WiLLM** to align heterogeneous CSI as a shared “language”; built a **259,774**-sample corpus spanning **12** datasets and **6** task categories
- Trained dataset-specific tokenizers with a shared GPT-style Transformer, reaching **91.06%** accuracy and robust **1%**-label few-shot adaptation

INTERNSHIP EXPERIENCE

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| <b>AGIQUAD</b>                                                                                                                                                                                                                                                                                                                                                                                                                              | May 2026 – Present    |
| <i>Embodied Intelligence Research Intern; supervised by Qiwei Xu; Vision-and-Language Navigation (VLN)</i><br>Shenzhen, China                                                                                                                                                                                                                                                                                                               |                       |
| <ul style="list-style-type: none"><li>Developing <b>MapFusion</b>, a query-conditioned VLN stack that grounds open-ended language in heterogeneous maps, reachable goals, and executable actions</li><li>Validated on an AgiBot <b>KuoTuo D2 Max</b> through cross-floor real-robot trials; extending toward inspection-oriented embodied applications and an <b>ICRA</b> submission</li></ul>                                              |                       |
| <b>Shanghai AI Laboratory</b>                                                                                                                                                                                                                                                                                                                                                                                                               | Jan. 2023 – Dec. 2024 |
| <i>Remote Research Intern; supervised by Xinyu Cai</i><br>Remote                                                                                                                                                                                                                                                                                                                                                                            |                       |
| <ul style="list-style-type: none"><li>Built an LLM-based traffic-scenario generator that converts natural-language requirements into <b>OpenSCENARIO</b> cases with generator-judge agents; configured CARLA–Apollo communication and Waymo-to-V2X-ViT conversion</li><li>Generated and quality-checked <b>InternVL</b> VLM-OCR data through difficult-sample synthesis, rule-based filtering, and parameter-freezing fine-tuning</li></ul> |                       |
| <b>Apple Inc.</b>                                                                                                                                                                                                                                                                                                                                                                                                                           | Jul. 2022 – Mar. 2023 |
| <i>Data Analyst Intern, Channel Planning and Forecast</i><br>Beijing, China                                                                                                                                                                                                                                                                                                                                                                 |                       |
| <ul style="list-style-type: none"><li>Built a modular Python/C++ allocation engine processing <b>1M+</b> weekly iPad units in approximately <b>2 minutes</b></li><li>Eliminated observed allocation errors in the evaluated workflow and reduced AOS lookup latency from <b>20s</b> to <b>5s</b>; automated pricing, reporting, and launch workflows</li></ul>                                                                              |                       |

PROJECT EXPERIENCE

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| <b>6G Network Digital Twin and Ray-Tracing Wireless Simulation Platform</b>                                                                                                                                                                                                                                                                         | Sept. 2024 – Present  |
| <i>Project participant, Guangdong Major Project on Environment-Enhanced 6G Networks</i><br>CUHK-Shenzhen                                                                                                                                                                                                                                            |                       |
| <ul style="list-style-type: none"><li>Contributed to a <b>Sionna RT</b>/Taichi RT digital-twin pipeline, diagnosing OSM/GPS alignment, ray/path extraction, benchmarking, and visualization</li><li>Designed semantic-aware calibration with LiDAR maps, SAM segmentation, sparse RSRP fitting, antenna sampling, and phase-aware pruning</li></ul> |                       |
| <b>Multi-Sensor UAV Docking and Adaptive Cruise Control System</b>                                                                                                                                                                                                                                                                                  | Oct. 2021 – Jan. 2023 |
| <i>Project Leader; supervised by Prof. Haiyuan Li</i>                                                                                                                                                                                                                                                                                               |                       |
| <ul style="list-style-type: none"><li>Led system design and integration for multi-sensor UAV aerial docking and adaptive cruise control; implemented PnP-based visual localization for centimeter-level docking</li><li>Coordinated algorithm validation and competition delivery; won university and Internet+ municipal third prizes</li></ul>    |                       |

TECHNICAL SKILLS

|                                            |                                              |                                                 |
|--------------------------------------------|----------------------------------------------|-------------------------------------------------|
| <b>Programming:</b> Python, C/C++, PyTorch | <b>Robotics:</b> ROS, CARLA, Apollo, MPC, A* | <b>Wireless:</b> Sionna RT, Taichi RT, CSI, SAM |
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AWARDS AND HONORS

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| Chinese Invention Patent (AFSP/ICRA-related): “Method and Related Device for Autonomous-Driving Path Planning Integrating Large-Model Reasoning,” ZL 2026 1 0345449.4. |           |
| Outstanding Undergraduate Thesis, Queen Mary-BUPT Joint Programme: WiFi CSI Action Recognition                                                                         | Jun. 2024 |
| The 8th China International “Internet+” College Student Innovation and Entrepreneurship Competition, BUPT Industrial Track Second Prize                                | Jul. 2022 |
| First Prize, National College Mathematics Competition                                                                                                                  | Jul. 2021 |

ENGLISH PROFICIENCY

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| <b>English:</b> TOEFL iBT 103/120 (R27/L28/S23/W25), CET-6 516, CET-4 607 |
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ACADEMIC SERVICE

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| <b>Reviewer:</b> ACL ARR 2026, ICRA 2025/2026, ICML 2026, IROS 2025, NeurIPS 2026; IEEE TMC, IEEE WCL, PIMRC 2025, GLOBECOM 2025 |
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